

Class 19: Pertussis Mini Project

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Background

Pertussis is a bacterial lung infection also known as whooping cough. Let’s begin by examining CDC reported case numbers in the US.

```
cdc <- data.frame(  
  year = c(1922L,1923L,1924L,1925L,  
           1926L,1927L,1928L,1929L,1930L,1931L,  
           1932L,1933L,1934L,1935L,1936L,  
           1937L,1938L,1939L,1940L,1941L,1942L,  
           1943L,1944L,1945L,1946L,1947L,  
           1948L,1949L,1950L,1951L,1952L,  
           1953L,1954L,1955L,1956L,1957L,1958L,  
           1959L,1960L,1961L,1962L,1963L,  
           1964L,1965L,1966L,1967L,1968L,1969L,  
           1970L,1971L,1972L,1973L,1974L,  
           1975L,1976L,1977L,1978L,1979L,1980L,  
           1981L,1982L,1983L,1984L,1985L,  
           1986L,1987L,1988L,1989L,1990L,  
           1991L,1992L,1993L,1994L,1995L,1996L,  
           1997L,1998L,1999L,2000L,2001L,  
           2002L,2003L,2004L,2005L,2006L,2007L,  
           2008L,2009L,2010L,2011L,2012L,  
           2013L,2014L,2015L,2016L,2017L,2018L,
```

```

                                2019L,2020L,2021L,2022L,2023L,2024L),
cases = c(107473,164191,165418,152003,
                                202210,181411,161799,197371,
                                166914,172559,215343,179135,265269,
                                180518,147237,214652,227319,103188,
                                183866,222202,191383,191890,109873,
                                133792,109860,156517,74715,69479,
                                120718,68687,45030,37129,60886,
                                62786,31732,28295,32148,40005,
                                14809,11468,17749,17135,13005,6799,
                                7717,9718,4810,3285,4249,3036,
                                3287,1759,2402,1738,1010,2177,2063,
                                1623,1730,1248,1895,2463,2276,
                                3589,4195,2823,3450,4157,4570,
                                2719,4083,6586,4617,5137,7796,6564,
                                7405,7298,7867,7580,9771,11647,
                                25827,25616,15632,10454,13278,
                                16858,27550,18719,48277,28639,32971,
                                20762,17972,18975,15609,18617,
                                6124,2116,3044,7063,22538)
)

```

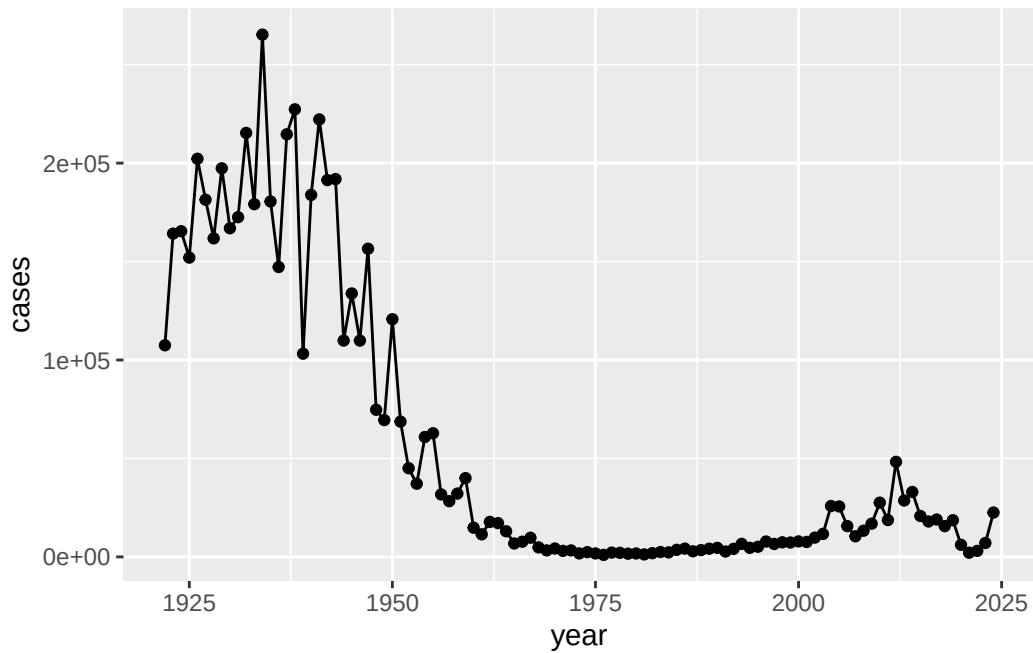
Plot of cases per year for pertussis in the US

```
library(ggplot2)
```

```

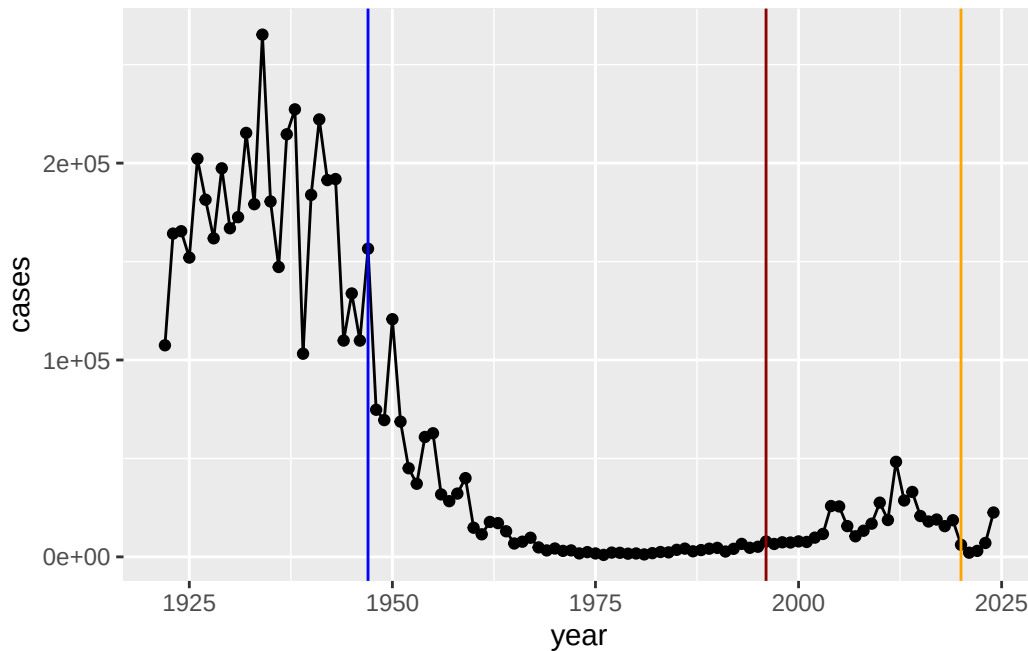
ggplot(cdc)+
  aes(year,
       cases)+
  geom_line()+
  geom_point()

```



Add some major milestone timeplints to our plot:

```
ggplot(cdc)+  
  aes(year,  
    cases)+  
  geom_line()+  
  geom_point()+  
  geom_vline(xintercept = 1947, col = "blue")+  
  geom_vline(xintercept = 1996, col = "darkred")+  
  geom_vline(xintercept = 2020, col = "orange")
```



The full introduction of mandatory wP (whole-cell) Pertussis immunization in the 1940s led to a dramatic reduction in case numbers (from over 200,000 to 100s).

The switch to the aP (newer acellular) formalization led to a slight increase in case numbers because of the anti-vaccine movement and the effectiveness of vaccine.

The COVID pandemic in 2020 led to a decrease in case numbers since people have more public health awareness.

The CMI-PB Project

The CMI-PB project aims to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

They make their data available via JSON format API endpoints- basically the database tables in a key: value type format like "infancy_vac": "wP". To read this we can use the `read_json()` function from the **jsonlite** package.

```
library(jsonlite)

subject <- read_json(path="https://www.cmi-pb.org/api/v5_1/subject",
                      simplifyVector = T)

head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. How many subject/individuals are in this dataset?

```
nrow(subject)
```

```
[1] 172
```

There are 172 subject/individuals in this dataset.

Q. How many wP and aP subjects are there

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

There are 87 aP subjects.

Q. What is the breakdown by “biological_sex” and “race”?

```
table(subject$biological_sex)
```

```
Female  Male
112     60
```

```
table(subject$race)
```

```

American Indian/Alaska Native
      1
Asian
     44
Black or African American
      5
More Than One Race
     19
Native Hawaiian or Other Pacific Islander
      2
Unknown or Not Reported
     21
White
     80

```

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

This breakdown is not particularly representative of the US population - this is a serious caveat for this study. However, it is still the largest sample of its type ever assembled.

```
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen",simplifyVector = T)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer",simplifyVector = T)
```

```
head(specimen)
```

```
specimen_id subject_id actual_day_relative_to_boost
```

1	1	1	-3
2	2	1	1
3	3	1	3
4	4	1	7
5	5	1	11
6	6	1	32

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

We need to “join” or link these tables with the `subject` table so we can begin to analyze this data and know who a given Ab sample was collected for and when.

```
library(dplyr)

meta <- inner_join(subject,specimen)
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White
4	1	wP	Female	Not Hispanic or Latino	White
5	1	wP	Female	Not Hispanic or Latino	White
6	1	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1
2	1986-01-01	2016-09-12	2020_dataset	2
3	1986-01-01	2016-09-12	2020_dataset	3
4	1986-01-01	2016-09-12	2020_dataset	4
5	1986-01-01	2016-09-12	2020_dataset	5
6	1986-01-01	2016-09-12	2020_dataset	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood

6		32		30	Blood
	visit				
1	1				
2	2				
3	3				
4	4				
5	5				
6	6				

Now let's join the `ab_titer` table with our 'meta' table so we have all information about a given measurement.

```
ab_data <- inner_join(ab_titer,meta)
```

Joining with ``by = join_by(specimen_id)``

```
head(ab_data)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	UG/ML	2.096133	1	wP	Female
2	IU/ML	29.170000	1	wP	Female
3	IU/ML	0.530000	1	wP	Female
4	IU/ML	6.205949	1	wP	Female
5	IU/ML	4.679535	1	wP	Female
6	IU/ML	2.816431	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood

2	-3	0	Blood
3	-3	0	Blood
4	-3	0	Blood
5	-3	0	Blood
6	-3	0	Blood

	visit
1	1
2	1
3	1
4	1
5	1
6	1

Q. how many Ab measurements do we have

We have 61956 measurements.

Q. How many different isotypes (types of Ab) are in the dataset

```
unique(ab_data$isotype)
```

```
[1] "IgE" "IgG" "IgG1" "IgG2" "IgG3" "IgG4"
```

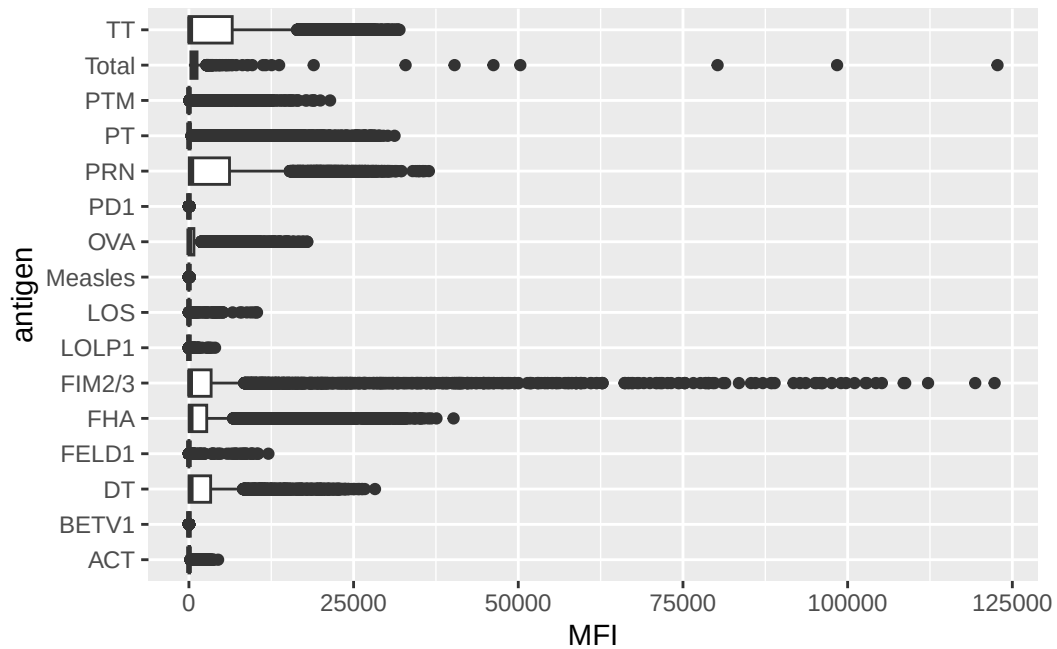
Q How many antigens are there

```
unique(ab_data$antigen)
```

```
[1] "Total" "PT" "PRN" "FHA" "ACT" "LOS" "FELD1"
[8] "BETV1" "LOLP1" "Measles" "PTM" "FIM2/3" "TT" "DT"
[15] "OVA" "PD1"
```

```
ggplot(ab_data)+
  aes(MFI,antigen)+
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range (`stat\_boxplot()`).

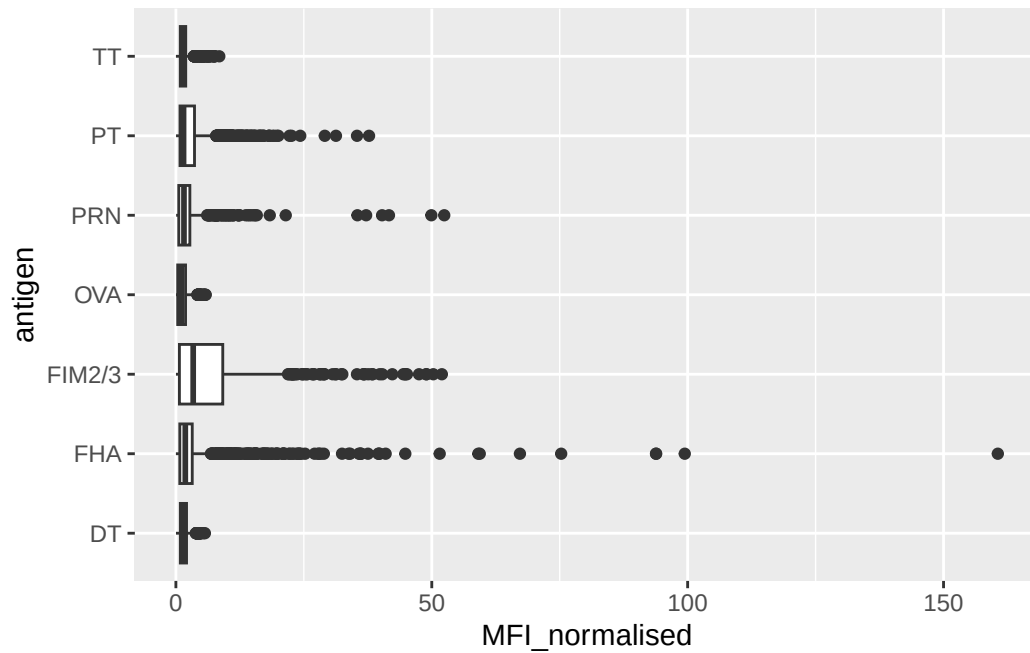


Examine IgG Ab titer levels

IgG is crucial for long-term immunity and responding to bacterial and viral infections.

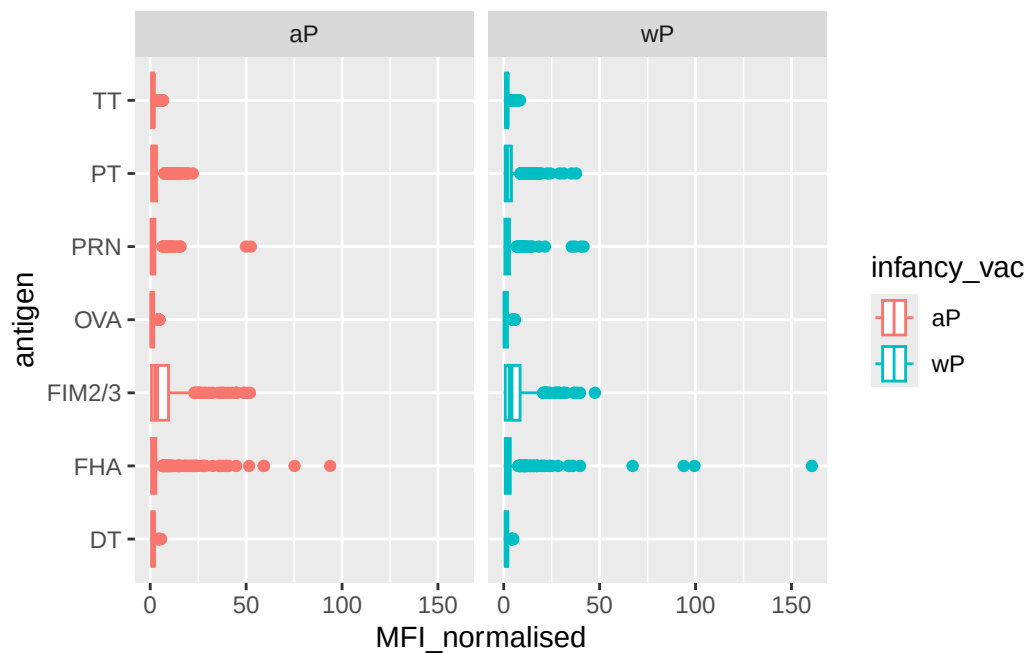
```
igg <- ab_data|>
  filter(isotype=="IgG")
```

```
ggplot(igg)+
  aes(MFI_normalised,antigen)+
  geom_boxplot()
```

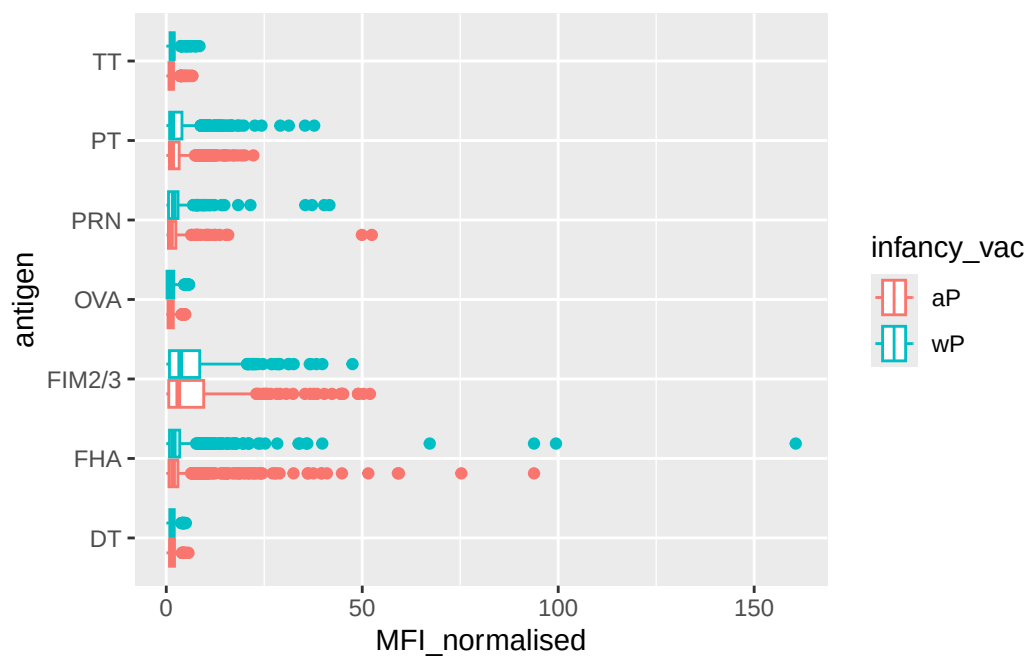


We can “facet” our plot by wP vs. aP

```
ggplot(igg)+
  aes(MFI_normalised,antigen, col=infancy_vac)+
  geom_boxplot()+
  facet_wrap(~infancy_vac)
```

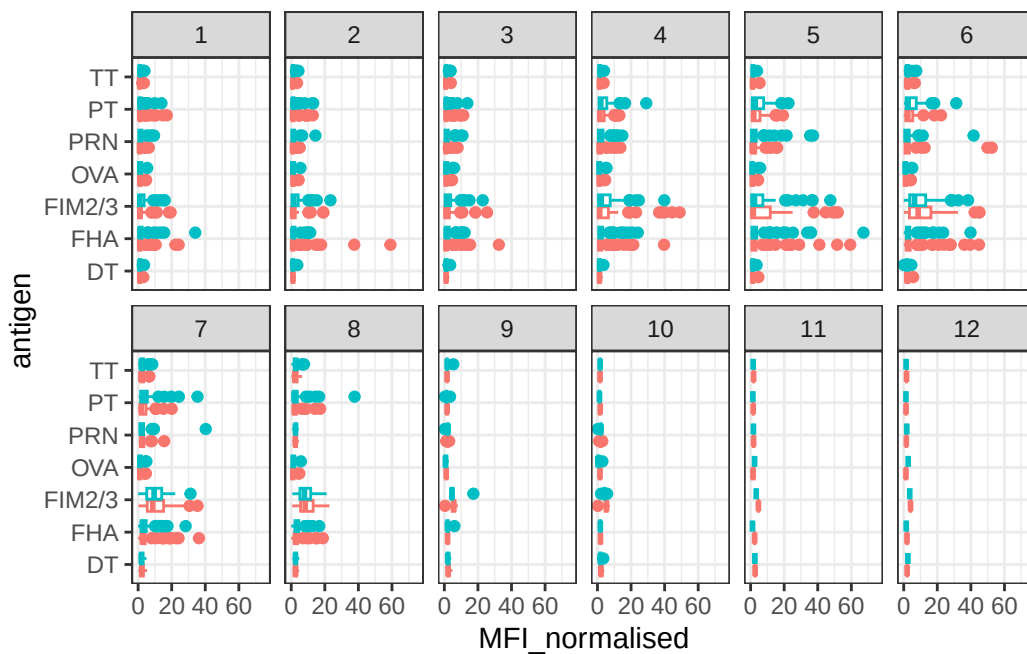


```
ggplot(igg)+
  aes(MFI_normalised,antigen, col=infancy_vac)+
  geom_boxplot()
```



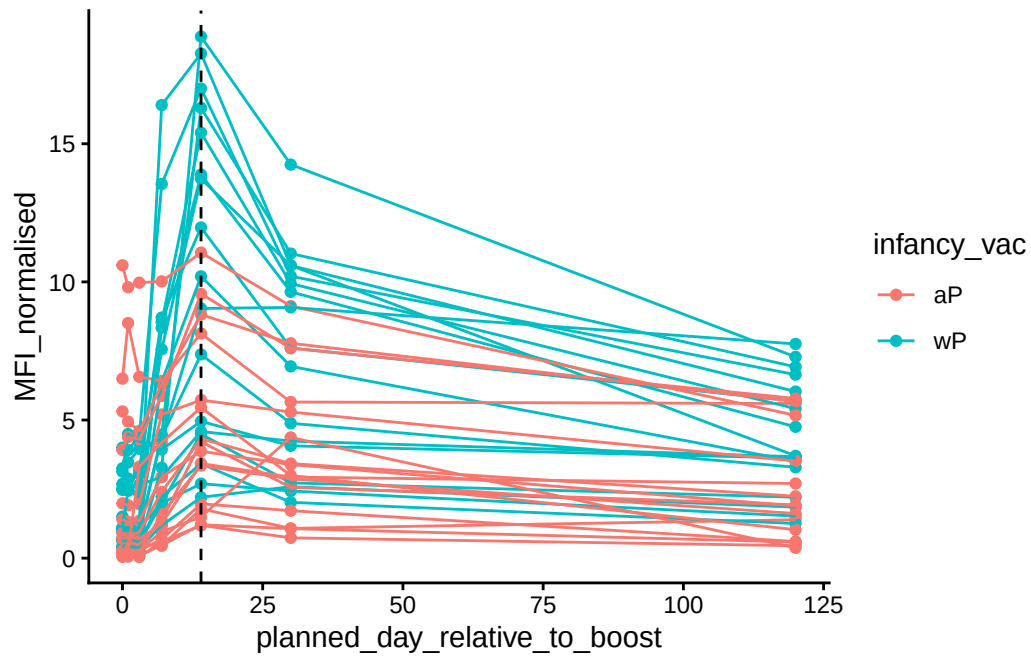
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



More advanced analysis digging into individual antigen responses over time:

```
filter(igg, antigen=="PT", dataset == "2021_dataset") |>
  ggplot() +
  aes(x=planned_day_relative_to_boost,
      y=MFI_normalised,
      col=infancy_vac,
      group=subject_id) +
  geom_point()+
  geom_line()+
  geom_vline(xintercept=14, linetype="dashed")+
  theme_classic()
```



The plot shows the time course of Pertussis toxin (PT) antibody responses for a large set of wP (teal) and aP (red) individuals. Levels peak at day 14 and are larger in magnitude for wP than aP individuals.

There are lots of cool things to explore in this dataset and we need coding and biology knowledge to do it effectively.